



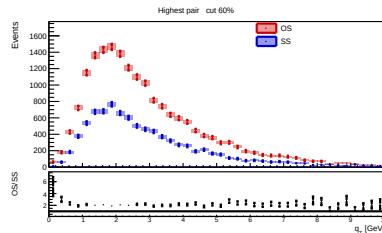
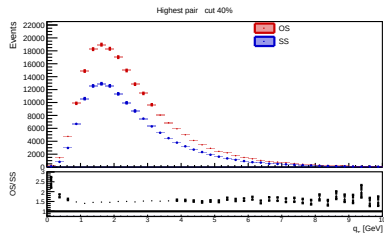
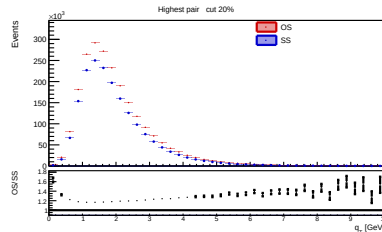
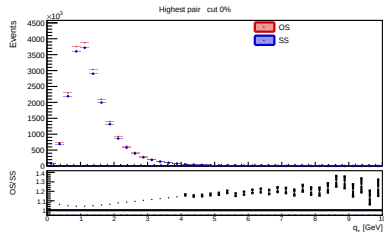
Shane Sweetman

*Project update meeting
Week 10*

A TMD-oriented analysis of $\pi^+\pi^-$ pairs in e^+e^- collisions

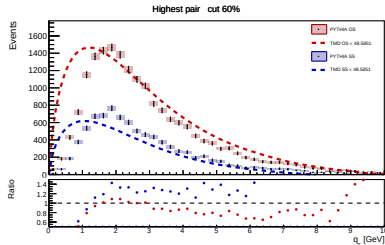
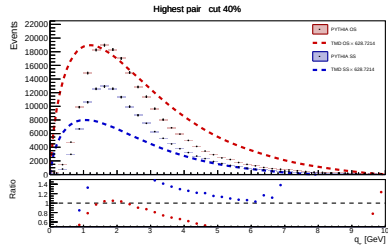
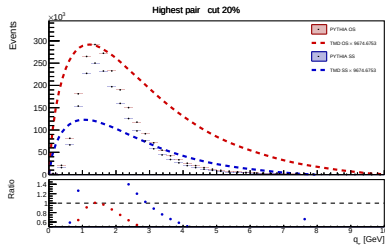
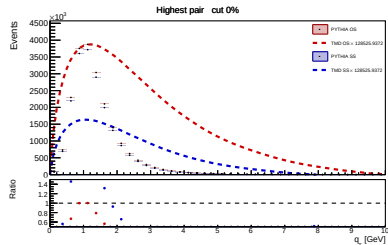
- › Fixed graphs.
- › TMD.
- › Graphs with $z = 0.7$.
- › Best- z scan.
- › Best % and best z .
- › Recap of best chi-square values.
- › Herwig (appendix).

Fixed graphs



- The last code was missing some of the selection cuts, which was what led to the small-width problem.
- This is now fixed in the updated version.
- Runtime was reduced by running multiple jobs at the same time with different random seeds.
- Higher cut % leads to bigger width.

Comparing with TMD theory



➤ Comparison with the TMD theory using the same rescaling for OS and SS.

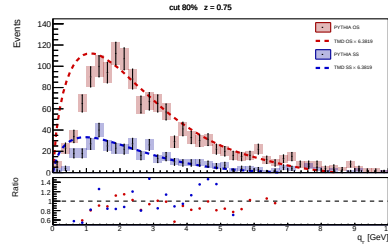
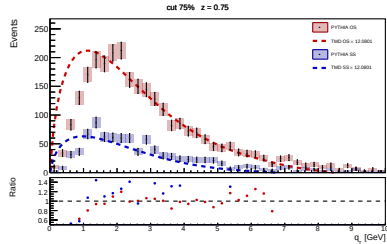
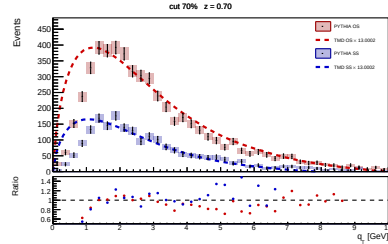
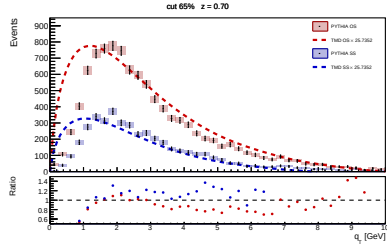
➤ The 60% cut starts to match reasonably well for $5+$ bins, i.e. above about 1.25 GeV.

➤ Using all bins, the chi-square is about 270.

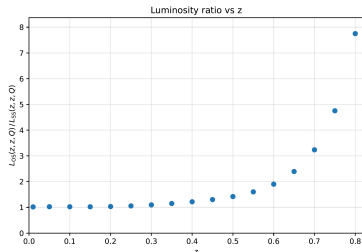
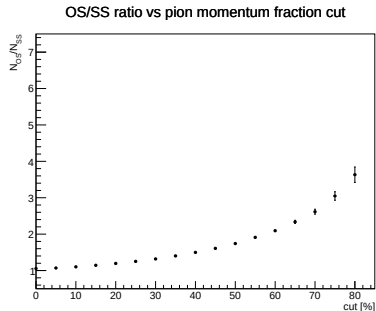
➤ Removing the first 5 bins reduces this to about 9, which is still not a good fit.

z scan to minimise chi-square

- Scan over z to minimise the chi-square.
- Shown for the 65%, 70%, 75%, and 80% cuts.
- Higher cuts were not taken, as the error bars become too large.



OS/SS ratio and luminosity formulas



$$\mathcal{L}_{OS}(b) = \sum_q B_q \left[D_q^{\pi^+}(z_1, b) D_{\bar{q}}^{\pi^-}(z_2, b) + D_{\bar{q}}^{\pi^+}(z_1, b) D_q^{\pi^-}(z_2, b) \right],$$

$$\mathcal{L}_{SS}(b) = \sum_q B_q \left[D_q^{\pi^+}(z_1, b) D_{\bar{q}}^{\pi^+}(z_2, b) + D_{\bar{q}}^{\pi^+}(z_1, b) D_q^{\pi^+}(z_2, b) \right].$$

- These luminosities are the theory objects used to represent the OS and SS pion-pair selections.
- The code is written in terms of π^+ fragmentation functions, with the π^- channel included through charge-conjugation relations.
- They therefore probe different combinations of favoured and unfavoured fragmentation.
- This supports the favoured vs unfavoured fragmentation picture.

2D PYTHIA parameter scan heatmap placeholder

Colour shows fit quality relative to default PYTHIA, with values below 1 giving better agreement.

- 7×7 scan of the PYTHIA parameters `StringPT:sigma` and `TimeShower:alphaSvalue`
- Done for the 60% pion momentum fraction cut using the fixed $z = 0.7$ theory comparison.
- Each square corresponds to one full 20 million event PYTHIA run with the same analysis cuts as before.

Questions?